

Restrictions of Infinite Circuits

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Abstract

We present a new approach to lower bounds on Borel rank via a connection between Borel sets and infinite circuits. We prove a sharp *restriction lemma*, which shows that by fixing the values of some inputs, we can simultaneously reduce the complexity of one Borel function while largely maintaining the complexity of another. This lemma provides a purely combinatorial proof of the Borel hierarchy theorem, as well as simple proofs of various Ramsey-like properties of Borel sets and functions. We also discuss counterexamples demonstrating the limitations of the restriction technique, in the context of both Borel functions and AC^0 circuit families.

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1 Introduction

Proving lower bounds on Borel rank relies on the Borel hierarchy theorem and its proof via universal sets and diagonalization. In this paper, we give a more concrete alternative approach based on a correspondence between Borel sets and infinite circuits. Along with proving the Borel hierarchy theorem, our techniques give simple proofs of several Ramsey-like properties of Borel sets and functions.

An infinite circuit is a countable, directed, well-founded graph with vertices labeled by $\wedge$ -gates, $\vee$ -gates, literals x_i , and negated literals $\bar{x}_i$. These circuits compute maps which take a string $x \in 2^{\mathbb{N}}$ as input and output the evaluation of some designated *output gate* to 0 or 1. The functions $f: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ computed by infinite circuits are precisely the characteristic functions of Borel sets $B \subseteq 2^{\mathbb{N}}$ (see Section 2.3 for a proof). Moreover, the Borel rank of B is closely related to the depth of the corresponding circuit.

Many classical theorems about Borel sets have analogous results for AC^0 , the class of functions $f: 2^{<\mathbb{N}} \rightarrow \{0, 1\}$ computable by polynomial-sized circuit families with bounded depth. The Borel hierarchy theorem corresponds to Sipser’s result that, for any $d \in \mathbb{N}$, there are functions computed by polynomial-sized circuits of depth $d + 1$ but not by polynomial-sized circuits of depth d . The fact that no infinite parity function $f: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ (a function whose output flips whenever a single input bit is flipped) is Borel corresponds to Furst, Saxe, and Sipser’s result that parity is not in AC^0 . Both of these theorems about finite circuits were originally proven using inspiration from the analogous infinitary result [Sip83; FSS81].

The sharpest versions of the AC^0 depth hierarchy theorem and parity lower bounds were proven by Håstad in [Hås86] using a theorem called the *switching lemma*. This result essentially states that, by fixing a reasonably small number of input bits to 0 or 1, one can dramatically simplify a circuit. Such a partial setting of the input bits is called a *restriction*, and we write f^ρ to denote the application of a restriction ρ to a function f .

In this paper, we prove an infinite analog of the switching lemma, which we call the *restriction lemma*. Our result concerns the γ -tree functions, which are computed by circuits obtained from labeling the vertices of a fully-branching tree of rank $\gamma < \omega_1$ with alternating $\wedge$ - and $\vee$ -gates (see Section 2.4 for a complete definition). These functions compute canonical

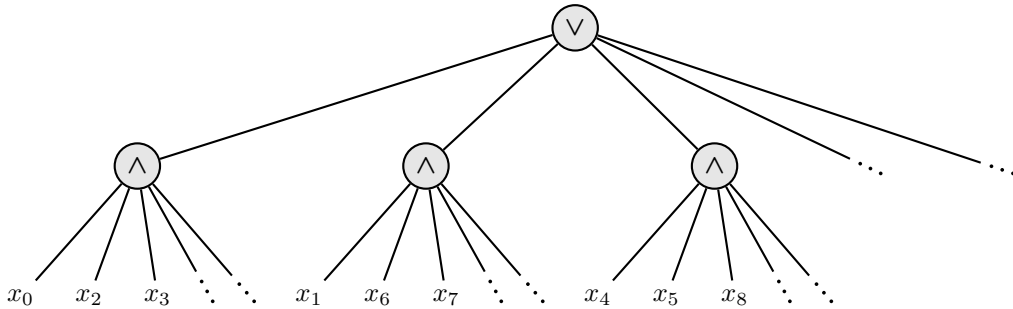


Figure 1: A circuit computing a 2-tree function. Each gate has countably infinite fan-in.

Σ_γ^0 - and Π_γ^0 -complete sets. The restriction lemma gives us a way to simultaneously reduce the complexity of a $\Sigma_{\beta+1}^0$ -measurable function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ and a $(\beta + \alpha)$ -tree function $g: 2^\mathbb{N} \rightarrow \{0, 1\}$ by applying a restriction. The precise statement of the theorem is as follows:

Theorem 1.1 (The restriction lemma). *For any ordinals $\alpha, \beta < \omega_1$, $\Sigma_{\beta+1}^0$ -measurable function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$, and $(\beta + \alpha)$ -tree function $g: 2^\mathbb{N} \rightarrow \{0, 1\}$, there exists a restriction ρ such that f^ρ is continuous and g^ρ is an α -tree function.*

When we are only concerned with Borel functions $f: 2^\mathbb{N} \rightarrow \{0, 1\}$ and $\alpha \geq 1$, we can choose ρ such that f^ρ is constant, rather than merely continuous:

Theorem 1.2 (Constant restriction lemma). *For any ordinals $\alpha, \beta < \omega_1$ with $\alpha \geq 1$, $\Sigma_{\beta+1}^0$ -measurable function $f: 2^\mathbb{N} \rightarrow \{0, 1\}$, and $(\beta + \alpha)$ -tree function $g: 2^\mathbb{N} \rightarrow \{0, 1\}$, there exists a restriction ρ such that f^ρ is constant and g^ρ is an α -tree function.*

While Sipser proved a version of these results for finite-depth circuits [Sip83], his techniques fundamentally break down at the transfinite level. Roughly speaking, he provides a construction which reduces the rank of f and g by 1, but iterating his argument ω -many times causes the complexity of g to completely collapse. Our proof circumvents these difficulties using a new inductive hypothesis which reduces the complexity of f and g by a possibly infinite amount. Our argument also requires a new technique for finding restrictions at limit stages of the induction. As was the case for Håstad's switching lemma in finite circuit complexity, the restriction lemma establishes a variety of results in descriptive set theory, including the Borel hierarchy theorem and the fact that no infinite parity function is Borel.

1.1 Outline of the paper

We begin with a review of Borel sets and functions in Section 2.1, and we define the notion of an infinite circuit in Section 2.2. We establish the key connection between Borel sets and circuits in Section 2.3. We define tree functions in Section 2.4 and restrictions (along with some important related terminology) in Section 2.5. Section 3 features the proofs of Theorem 1.1 and Theorem 1.2.

We explore the consequences of the restriction lemma in Section 4, beginning with a new proof of the Borel hierarchy theorem in Section 4.1. We prove in Section 4.2 that any Borel function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ can be made continuous by applying a restriction preserving infinitely many variables, and we use this to show that no infinite parity function is Borel in Section 4.3. We also give a generalization of the restriction lemma to arbitrary Borel functions in Section 5, rather than just γ -tree functions. This requires us to broaden the class of restrictions we allow, and we show that the theorem fails without this relaxation. We also discuss similar limitations in applying restrictions to functions in AC^0 .

We prove the sharpness of the restriction lemma in its strongest form (Theorem 1.1) in Section 6, and we discuss how the weaker form (Theorem 1.2) is not sharp. We introduce an open conjecture regarding how far Theorem 1.2 is from optimal, and we provide some partial progress towards a resolution.

2 Borel sets and infinite circuits

Define $2^{<\mathbb{N}} = \bigcup_{n \in \mathbb{N}} \{0, 1\}^n$, and let $w_0 w_1 \dots w_{k-1} w_k$ denote the concatenation of strings $w_0, w_1, \dots, w_{k-1} \in 2^{<\mathbb{N}}$ and $w_k \in 2^{<\mathbb{N}} \cup 2^{\mathbb{N}}$. We consider the *Cantor space* $2^{\mathbb{N}}$, equipped with the topology generated by basic open sets of the form $w2^{\mathbb{N}}$ for $w \in 2^{<\mathbb{N}}$.

2.1 Preliminaries

We first recall some standard definitions from descriptive set theory about Borel sets and functions. See [Kec95] for a detailed treatment.

Definition 2.1. A set $S \subseteq 2^{\mathbb{N}}$ is called Σ_0^0 or Π_0^0 if it is clopen. A set is Σ_α^0 (resp. Π_α^0) if it is a countable union (resp. intersection) of sets S_n for $n \in \mathbb{N}$, each of which is in $\Pi_{\beta_n}^0$ (resp. $\Sigma_{\beta_n}^0$) for some $\beta_n < \alpha$. We call a set Δ_α^0 if it is simultaneously Σ_α^0 and Π_α^0 .

Remark 2.2. This definition agrees with the usual definition of the Borel hierarchy since the Cantor space is zero-dimensional (has a clopen basis). In particular, the open (resp. closed) sets of $2^{\mathbb{N}}$ are precisely the countable unions (resp. intersections) of clopen sets.

Definition 2.3. A set $S \subseteq 2^{\mathbb{N}}$ is *Borel* if it is Σ_α^0 for some $\alpha < \omega_1$. We let $\mathcal{B}$ denote the class of Borel sets. The *rank* of a Borel set B is the smallest α such that $B \in \Sigma_\alpha^0 \cup \Pi_\alpha^0$.

Definition 2.4. Given a class $\mathcal{C} \subseteq \mathcal{P}(2^{\mathbb{N}})$, we call a function $f: 2^{\mathbb{N}} \rightarrow Y$ (where $Y = 2^{\mathbb{N}}$ or $\{0, 1\}$) *$\mathcal{C}$ -measurable* if $f^{-1}(S) \in \mathcal{C}$ whenever $S \subseteq Y$ is open.

Definition 2.5. A function $f: 2^{\mathbb{N}} \rightarrow Y$ is *Borel* if it is $\mathcal{B}$ -measurable.

Remark 2.6. The facts that $2^{\mathbb{N}}$ has a countable basis and that Σ_α^0 is closed under countable unions can be used to show that every Borel function is Σ_α^0 -measurable for some $\alpha < \omega_1$.

Remark 2.7. Suppose B is a Borel set with rank α . Then the characteristic function of B is $\Sigma_{\alpha+1}^0$ -measurable, and it is Σ_α^0 -measurable if and only if $B \in \Delta_\alpha^0$.

Definition 2.8. Given a class $\mathcal{C}$, we call a set $S \subseteq 2^{\mathbb{N}}$ *$\mathcal{C}$ -hard* if for every $S' \in \mathcal{C}$, there exists a continuous map $\varphi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that $x \in S' \iff \varphi(x) \in S$. We call φ a *continuous reduction*. We call a $\mathcal{C}$ -hard set $S \subseteq 2^{\mathbb{N}}$ *$\mathcal{C}$ -complete* if $S \in \mathcal{C}$.

2.2 Infinite circuits

Definition 2.9. A *circuit* C on variables $x_0, x_1, \dots$ is a countable, labeled, and directed graph. We call the vertices of C *gates*, and if there is an edge from a gate G' to a gate G , we say that G' is a *child* of G . A circuit must be well-founded under the child relation. Any gate with fan-in (i.e. in-degree) at least 1 is called an *internal gate* and must be labeled with either the symbol $\wedge$ or $\vee$. The remaining nodes are called *input gates* and must be labeled with either 0, 1, or the symbol x_i or $\bar{x}_i$ for some $i \in \mathbb{N}$ (denoting either the i -th bit of x or the negation of this bit). We call the labels 0, 1, $x_i, \bar{x}_i$ *literals*.

Definition 2.10. Given a gate G in C and some input $x \in 2^{\mathbb{N}}$, we define the *evaluation* $G(x)$ inductively (which we may do by well-foundedness). An input gate evaluates to its label, an $\wedge$ -gate evaluates to the conjunction of the evaluations of its children, and an $\vee$ -gate evaluates to the disjunction of the evaluations of its children. One vertex G_{out} of C is designated as the *output*, and we write $C(x) = G_{\text{out}}(x)$. Thus C computes a function from $2^{\mathbb{N}}$ to $\{0, 1\}$. We call two circuits *equivalent* if they compute the same function.

Definition 2.11. We say that a function $f: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ *computes* the set $f^{-1}(\{1\})$.

Remark 2.12. Allowing for $\neg$ -gates would not increase the power of these circuits, since De Morgan's law allows us to propagate all negations to the bottom layer of the circuit.

Definition 2.13. We call a circuit C *alternating* if no $\vee$ -gate in C has another $\vee$ -gate as a child and no $\wedge$ -gate in C has another $\wedge$ -gate as a child.

Remark 2.14. By merging non-alternating gates, we see that every circuit is equivalent to an alternating circuit.

Definition 2.15. Let G be a gate of a circuit C . The *subcircuit of C rooted at G* is the subgraph of C induced by G and its descendants (where the descendant relation is the transitive closure of the child relation), with G designated as the output gate.

Definition 2.16. A *child* of a circuit C is a subcircuit of C rooted at a child of the output gate of C .

Definition 2.17. We call a circuit Σ_0 or Π_0 if it is finite. We call a circuit C a Σ_α -circuit (resp. Π_α -circuit) if its output gate is an $\vee$ (resp. $\wedge$)-gate and, for each child C' of C , there exists $\beta < \alpha$ such that C' is a Π_β (resp. Σ_β)-circuit.

Definition 2.18. The *rank* of an alternating circuit C is the least ordinal α such that C is a Σ_α - or Π_α -circuit.

Remark 2.19. Remark 2.14 tells us that every circuit is equivalent to a rank- α circuit for some $\alpha < \omega_1$. By inductively duplicating shared subcircuits, we see that every rank- α circuit is equivalent to a rank- α circuit whose underlying graph is a tree.

2.3 Equivalence of Borel sets and circuits

Our similar choice of notation between Borel sets and circuits is deliberate, as these notions correspond precisely. The following result and its corollaries allow us to describe Borel subsets of $2^{\mathbb{N}}$ using circuits.

Lemma 2.20. *A set $S \subseteq 2^{\mathbb{N}}$ is clopen if and only if it is computed by a finite circuit.*

Proof. Any open set $S \subseteq 2^{\mathbb{N}}$ can be written as a union of basic open sets, and if S is also closed, this union can be taken to be finite by compactness. Each basic open set in such a union is computed by a finite circuit, so the forward direction follows by taking a finite disjunction of such circuits.

Conversely, any circuit consisting of a single input gate computes either $\emptyset$, $2^{\mathbb{N}}$, or a set of the form $\{x \in 2^{\mathbb{N}} : x_i = b\}$ for some $i \in \mathbb{N}$ and $b \in \{0, 1\}$. Note that these sets are all clopen. As the clopen sets are closed under finite unions and intersections, this means that any set S computed by a finite circuit is also clopen. $\square$

Corollary 2.21. *For $\alpha < \omega_1$, a set $S \subseteq 2^{\mathbb{N}}$ is Σ_α^0 (resp. Π_α^0) if and only if it is computed by a Σ_α (resp. Π_α)-circuit.*

Corollary 2.22. *A subset of $2^{\mathbb{N}}$ is Borel if and only if it is computed by a circuit.*

We can also characterize Borel functions using circuits:

Lemma 2.23. *A function $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ is Σ_α^0 -measurable if and only if the map $x \mapsto f(x)_i$ is computed by both a Σ_α -circuit and Π_α -circuit for every $i \in \mathbb{N}$.*

Proof. Fix $\alpha < \omega_1$, and suppose first that $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ is Σ_α^0 -measurable. Then the sets $f^{-1}(\{x \in 2^{\mathbb{N}} : x_i = b\})$ are Σ_α^0 for every $i \in \mathbb{N}$ and $b \in \{0, 1\}$, since the sets $\{x \in 2^{\mathbb{N}} : x_i = b\}$ are all open. Corollary 2.21 then proves the forward direction.

Conversely, suppose each map $x \mapsto f(x)_i$ for $i \in \mathbb{N}$ is computed by both a Σ_α -circuit and Π_α -circuit. Then the sets $f^{-1}(\{x \in 2^{\mathbb{N}} : x_i = b\})$ lie in Σ_α^0 for every $i \in \mathbb{N}$ and $b \in \{0, 1\}$. Every open set can be generated by basic open sets of the form $\{x \in 2^{\mathbb{N}} : x_i = b\}$ via countable unions and finite intersections, and as Σ_α^0 is closed under countable unions and finite intersections, this proves the claim. $\square$

Lemma 2.20 and Lemma 2.23 give us the following corollary:

Corollary 2.24. *A function $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ is continuous if and only if the map $x \mapsto f(x)_i$ is computed by a finite circuit for every $i \in \mathbb{N}$.*

2.4 Tree functions

The *tree functions* are a special family of Borel functions from $2^{\mathbb{N}}$ to $\{0, 1\}$ which compute Σ_α^0 - or Π_α^0 -complete sets. We will use these functions in the formulation of the restriction lemma, and they are defined in terms of circuits. We begin by defining the underlying tree structure of these circuits.

Definition 2.25. The *full trees* of rank $\alpha < \omega_1$, denoted T_α , are the rooted trees defined recursively as follows:

- The tree T_0 consists of a single vertex, which is the root node.
- For any successor ordinal $\alpha + 1$, the tree $T_{\alpha+1}$ consists of ω -many disjoint copies of T_α , all of whose root nodes are connected to a single additional vertex. This additional vertex is the root node of $T_{\alpha+1}$.
- If α is a limit ordinal, the tree T_α consists of ω -many disjoint copies of T_β for each $\beta < \alpha$, all of whose root nodes are connected to a single additional vertex. This additional vertex is the root node of T_α .

Definition 2.26. A *full alternating tree* of rank α is an alternating circuit whose underlying graph is T_α , whose input gates are injectively labeled by literals x_i , and whose output gate is the root of T_α .

Remark 2.27. Every full alternating tree of rank α is either a Σ_α - or Π_α -circuit, depending on whether the output gate is an $\vee$ -gate or an $\wedge$ -gate.

Definition 2.28. A function $g: 2^\mathbb{N} \rightarrow \{0, 1\}$ is called an α -*tree function* if it is computed by a full alternating tree of rank α .

Remark 2.29. One can construct a reduction from any given Borel set to the appropriate full alternating tree to see that every α -tree function defines either a Σ_α^0 - or Π_α^0 -complete set, respectively.

Remark 2.30. One can modify the definition of full trees at limit ordinals α to merely contain a sequence of disjoint full trees $T_{\beta_0}, T_{\beta_1}, \dots$ satisfying $\beta_n < \alpha$ for each $n \in \mathbb{N}$ and $\sup_{n \in \mathbb{N}} \beta_n = \alpha$. The results of our paper remain true, roughly because these trees all embed into each other. Our particular definition is chosen to make the proof of Lemma 3.1 easier.

2.5 Restrictions

We now define the key concept behind the restriction lemma:

Definition 2.31. A *restriction* on $2^\mathbb{N}$ is a function $\rho: \mathbb{N} \rightarrow \{0, 1, *\}$. We call variables in the set $\{x_i : \rho(i) = *\}$ *preserved* and the remaining variables *fixed*.

Definition 2.32. Given a function $f: 2^\mathbb{N} \rightarrow Y$ (where Y is equal to either $2^\mathbb{N}$ or $\{0, 1\}$) and restriction ρ , we define the *restricted function* $f^\rho: 2^{\rho^{-1}(\{*\})} \rightarrow Y$ which takes values for the preserved variables as input and evaluates f on the resulting setting.

Definition 2.33. Given a circuit C and restriction ρ , we define the *restricted circuit* C^ρ by replacing the labels x_i with $\rho(i)$ and $\bar{x}_i$ with $\overline{\rho(i)}$ whenever x_i is fixed.

Definition 2.34. We call a circuit C^ρ *forced* by ρ if it is equivalent to a constant 0- or 1-circuit. We call C^ρ *suppressed* if its output gate is an $\vee$ -gate and it is forced to 1 or if its output gate is an $\wedge$ -gate and it is forced to 0. A forced circuit is *overriding* if it is not suppressed. We call C^ρ *preserved* if it computes a 0-tree function.

Remark 2.35. In an alternating circuit, suppressed children do not affect the value of their parents and overriding children force their parents. Notice also that any full alternating rank-1 tree can be suppressed by assigning a single literal and can only be made overriding if all literals are assigned. Any circuit computing a non-constant function can be preserved by assigning one of its input variables to $*$ and the rest to 0 or 1.

3 Proof of the restriction lemma

Our goal in this section is to prove Theorem 1.1 and Theorem 1.2, and this requires us to view f through the lens of circuits. In particular, Lemma 2.23 tells us that a $\Sigma_{\beta+1}^0$ -measurable function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ corresponds to a countable collection of sets computable

both by $\Sigma_{\beta+1}$ - and $\Pi_{\beta+1}$ -circuits, and Corollary 2.24 tells us that a continuous function $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ corresponds to a countable collection of finite circuits. Instead of proving the lemma directly, we will prove a related claim more amenable to induction and rephrased in the language of circuits.

Lemma 3.1 (Inductive restriction lemma). *For any ordinal $\alpha < \omega_1$, countable collection of full rank- α alternating trees $(T_{ij})_{i,j \in \mathbb{N}}$ defined on disjoint sets of input variables, and countable collection $(C_n)_{n \in \mathbb{N}}$ of Σ_α - or Π_α -circuits defined on all of the input variables, there exists a restriction ρ such that each C_n^ρ is equivalent to a finite circuit, each T_{ij}^ρ is suppressed or preserved, and infinitely many trees T_{ij}^ρ are preserved for each fixed $i \in \mathbb{N}$.*

The collection $(T_{ij})_{i,j \in \mathbb{N}}$ should be interpreted as infinitely many groups of infinitely many alternating trees. The inductive restriction lemma tells us that no matter how these groups are formed, we can always ensure that each group $(T_{ij})_{j \in \mathbb{N}}$ for fixed $i \in \mathbb{N}$ contains infinitely many preserved trees.

3.1 Tree decomposition

To prove the restriction lemma, we need some information about the combinatorial structure of the full trees, and the lemma we prove in this section is the necessary result. We first recall two crucial facts about ordinal arithmetic (see [Jec02] for a detailed treatment):

Proposition 3.2 (Left subtraction). Given ordinals α and β with $\beta \leq \alpha$, there exists a unique ordinal γ such that $\beta + \gamma = \alpha$.

Proposition 3.3 (Right monotonicity). Given ordinals β, γ and δ , we have $\beta + \delta < \beta + \gamma$ if and only if $\delta < \gamma$.

We also provide some definitions and notation to state our combinatorial lemma. For the following three definitions, S and T are rooted trees.

Definition 3.4. We write $S \leq T$ if S is a subtree of T with the same root node as T .

Definition 3.5. A *child* of T is a subtree of T rooted at a child of the root node of T .

Definition 3.6. We define the *grafted* tree $S + T$ by replacing the leaves of T by disjoint copies of S (each leaf becomes the root node of a copy of S). The root of $S + T$ is the root of T .

A simple transfinite induction verifies that $\text{rank}(S + T) = \text{rank}(S) + \text{rank}(T)$. Our key lemma is the following property of the full trees:

Lemma 3.7. *We have whenever $\beta + \gamma = \alpha$ that $T_\beta + T_\gamma \leq T_\alpha$.*

Proof. We proceed by transfinite induction on γ . If $\gamma = 0$, then the result is immediate as $T_\beta + T_\gamma = T_\beta = T_\alpha$. For the successor step, suppose $\gamma = \delta + 1$ and that the claim holds for δ . Then $\alpha = \beta + \delta + 1$ is a successor ordinal, so T_α has ω -many $T_{\beta+\delta} \geq T_\beta + T_\delta$ trees as

children. We similarly have that T_γ has ω -many T_δ trees as children, so $T_\beta + T_\gamma$ has ω -many $T_\beta + T_\delta$ trees as children. Thus $T_\beta + T_\gamma \leq T_\alpha$, proving the successor case.

Now suppose γ is a limit ordinal and the claim holds for every $\delta < \gamma$. Then T_γ has ω -many copies of T_δ for each $\delta < \gamma$ as children. Thus $T_\beta + T_\gamma$ contains as children ω -many copies of the trees $T_\beta + T_\delta \leq T_{\beta+\delta}$ for each $\delta < \gamma$. Since each $\beta + \delta < \beta + \gamma = \alpha$ by right monotonicity, the tree T_α contains ω -many copies of each $T_{\beta+\delta}$ as children. Thus $T_\beta + T_\gamma \leq T_\alpha$, proving the limit case. $\square$

3.2 Proof of the inductive lemma

We are now ready to prove the restriction lemma in both forms (Theorem 1.1 and Theorem 1.2). First, let us temporarily assume Lemma 3.1 to be true and see how it implies our desired results.

Proof of Theorem 1.1. If $\alpha = 0$, the claim is immediate. Otherwise, Lemma 2.23 tells us that the sets $f^{-1}(\{x \in 2^\mathbb{N} : x_i = 1\})$ are computed both by $\Sigma_{\beta+1}$ -circuits C_i and $\Pi_{\beta+1}$ -circuits D_i for each $i \in \mathbb{N}$. Lemma 3.7 tells us that $T_\beta + T_\alpha \leq T_{\beta+\alpha}$, and we can find a restriction $\tilde{\rho}$ so that each $g^{\tilde{\rho}}$ is computed by an alternating circuit obtained from the tree $T_\beta + T_\alpha$. Apply Lemma 3.1 to every child of the C_i 's and D_i 's and to the T_β subtrees of $T_\beta + T_\alpha$, where each group consists of all T_β subtrees corresponding to leaves in T_α with the same parent. Letting ρ denote this further restriction, we see that each C_i^ρ is a Σ_1 -circuit, each D_i^ρ is a Π_1 -circuit, and g^ρ is an α -tree function. Lemma 2.23 tells us that f^ρ is Σ_1^0 -measurable and hence continuous. $\square$

Proof of Theorem 1.2. Define $\tilde{f}: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ so that $\tilde{f}(x)_0 = f(x)$ and $\tilde{f}_i(x) = 0$ for all $x \in 2^\mathbb{N}$ and $i \in \mathbb{N} \setminus \{0\}$. Theorem 1.1 then gives us a restriction $\tilde{\rho}$ for which $g^{\tilde{\rho}}$ is computed by a full alternating tree of rank $\alpha \geq 1$ and $(\tilde{f})^{\tilde{\rho}}$ is continuous. Corollary 2.24 tells us that $f^{\tilde{\rho}}$ depends on only finitely many variables $x_{i_0}, \dots, x_{i_k}$. By fixing each of these variables such that their corresponding input gates are suppressed, we obtain a further restriction ρ for which f^ρ is constant and g^ρ is still an α -tree function. $\square$

It remains to prove Lemma 3.1, and we proceed by transfinite induction on α . We can take ρ to preserve all variables in the $\alpha = 0$ case. We first give a proof of the lemma when $\alpha = 1$, since it will be helpful for both the inductive and limit stages of the induction. The proof when $\alpha = 1$ is similar to Sipser's argument in [Sip83].

Proof of the $\alpha = 1$ case. We are given a collection $(T_{ij})_{i,j \in \mathbb{N}}$ of full rank-1 alternating trees defined on disjoint sets of inputs, each of which contains a single $\wedge$ - or $\vee$ -gate with countably infinite fan-in, as well as Σ_1 - or Π_1 -circuits $(C_n)_{n \in \mathbb{N}}$ defined on all of the input variables. Our goal is to find a restriction ρ such that each T_{ij}^ρ is suppressed or preserved, infinitely many T_{ij}^ρ 's are preserved for each fixed i , and each C_n^ρ is equivalent to a finite circuit.

The challenge of constructing ρ is that we need to assign infinitely many $*$'s in order to ensure that infinitely many T_{ij}^ρ 's are preserved for each fixed i , but not in a manner that

makes any fixed C_n^ρ depend on infinitely many $*$'s. To address this difficulty, we define the *queue sequence* $(q_n)_{n \in \mathbb{N}} = (0, 0, 1, 0, 1, 2, 0, 1, 2, 3, \dots)$ so that every natural number appears infinitely many times in the sequence. We define ρ in stages, assigning one $*$ at each stage and ensuring that after the n -th stage, the circuit C_n depends only on the $n + 1$ variables preserved so far. In particular, we do the following for $n = 0, 1, 2, \dots$:

1. Preserve some previously untouched $T_{q_n, j}$ by preserving one of its inputs and fixing the remaining inputs.
2. For each of the 2^{n+1} possible settings of the preserved variables to 0 or 1, do the following:
 - (a) If C_n is unforced under this setting, it must have an unforced Σ_0/Π_0 child. Force this child to be overriding by setting finitely many new variables to 0 or 1.
 - (b) For each of the finitely many trees T_{ij} containing a newly assigned input variable, fix all other input variables of T_{ij} such that T_{ij} is suppressed.

Let $\tilde{\rho}$ denote the resulting partial restriction. Each $C_n^{\tilde{\rho}}$ depends only on the $n + 1$ bits preserved in steps 0 through n and is hence equivalent to a finite circuit. We also have that infinitely many T_{ij} 's are preserved for each fixed i . By assigning all remaining literals so that every other T_{ij} is suppressed or preserved, we get our desired restriction ρ . $\square$

Proof of the successor case. Let $(T_{ij})_{i, j \in \mathbb{N}}$ be full rank- $(\alpha + 1)$ alternating trees defined on disjoint sets of input variables and $(C_n)_{n \in \mathbb{N}}$ be $\Sigma_{\alpha+1}$ - or $\Pi_{\alpha+1}$ -circuits defined on all inputs. We apply the inductive hypothesis to the collection of all children of the T_{ij} 's and to the collection of all Σ_α - or Π_α -children of the C_n 's. We group the rank- α children of the T_{ij} 's by their parent so that each T_{ij} has infinitely many preserved children. We therefore have for the resulting restriction $\tilde{\rho}$ that each $T_{ij}^{\tilde{\rho}}$ is equivalent to a full rank-1 alternating tree and each $C_n^{\tilde{\rho}}$ is equivalent to a Σ_1 - or Π_1 -circuit. The $\alpha = 1$ case then gives us a further restriction ρ satisfying the desired properties. $\square$

At this point, we have proven the restriction lemma for finite α , which in particular suffices to prove the Borel hierarchy theorem for finite-rank Borel sets. This matches Sipser's result in [Sip83]. To obtain the full result, we just need to prove the limit case. Our strategy will be similar in spirit to the $\alpha = 1$ case, but instead of repeatedly reducing Σ_1 - or Π_1 -circuits to finite circuits by setting finitely many literals, we will repeatedly apply the inductive hypothesis to reduce entire Σ_β - or Π_β -circuits (with $\beta < \alpha$) to finite circuits.

Proof of the limit case. Suppose α is a limit ordinal, let $(T_{ij})_{i, j \in \mathbb{N}}$ be full rank- α alternating trees with disjoint input sets, and suppose $(C_n)_{n \in \mathbb{N}}$ are Σ_α - or Π_α -circuits defined on all inputs. Let $(D_k)_{k \in \mathbb{N}}$ denote the children of these circuits. We write $S_{ij0}, S_{ij1}, \dots$ to denote the children of the T_{ij} 's. We construct ρ in stages while maintaining the invariant that

$$\text{rank}(D_k) < \sup_{\ell \in \mathbb{N}} (\text{rank}(S_{ij\ell}))$$

for all $i, j, k \in \mathbb{N}$, which evidently holds before applying any restriction since the right hand side is equal to α . Define the queue sequence $(q_s, r_s)_{s \in \mathbb{N}}$ so that every $(q, r) \in \mathbb{N}^2$ appears infinitely many times in the sequence, and do the following for $s = 0, 1, 2, \dots$:

1. Choose a subtree $S_{q_s r_s \ell}$ which has not been yet forced or preserved and permanently preserve it, assigning a $*$ to a single variable x_{p_s} . This “permanent” assignment means that we cannot change the $*$ to a 0 or 1 in any subsequent step.
2. Let $\beta_s = \text{rank}(D_s)$, and suppress all subtrees $S_{ij\ell}$ such that $\text{rank}(S_{ij\ell}) < \beta_s$. The invariant ensures that infinitely many subtrees of each T_{ij} are not suppressed.
3. For each unforced $S_{ij\ell}$, let $\gamma_{ij\ell}$ denote the unique ordinal such that $\beta_s + \gamma_{ij\ell} = \text{rank}(S_{ij\ell})$. Using Lemma 3.7, we can fix some inputs of $S_{ij\ell}$ so that $S_{ij\ell}$ is equivalent to an alternating circuit defined on the tree $T_{\beta_s} + T_{\gamma_{ij\ell}}$.
4. Let $\mathcal{F}$ denote the family of all subcircuits of the D_k ’s with rank at most β_s . Note that $D_s \in \mathcal{F}$.
5. Replace each $C \in \mathcal{F}$ with the Σ_{β_s+1} -circuit

$$\bigvee_{\sigma_0 \dots \sigma_s \in \{0,1\}^{s+1}} \left(C^{\sigma_0 \dots \sigma_s} \wedge \bigwedge_{m=0}^s x_{p_m} = \sigma_m \right),$$

where $C^{\sigma_0 \dots \sigma_s}$ denotes the circuit C with each permanently preserved literal x_{p_m} set to σ_m .

6. Now apply the inductive hypothesis to all of the circuits $C^{\sigma_0 \dots \sigma_s}$ from the previous step and to the subtrees T_{β_s} , where each group consists of all T_{β_s} subtrees of a given tree $T_{\beta_s} + T_{\gamma_{ij\ell}}$ corresponding to leaves with the same parent. Let ρ_s denote the restriction so obtained.
7. Replace all circuits and trees with their equivalent simplified versions under ρ_s .

We make a few observations about the resulting circuit at the end of each step. First, since each $(C^{\sigma_0 \dots \sigma_s})^{\rho_s}$ depends on only finitely many preserved literals, the circuit

$$\bigvee_{\sigma_0 \dots \sigma_s \in \{0,1\}^{s+1}} \left(C^{\sigma_0 \dots \sigma_s} \wedge \bigwedge_{m=0}^s x_{p_m} = \sigma_m \right)^{\rho_s}$$

also depends on only finitely many preserved or permanently preserved literals and is equivalent to a finite circuit. Each restricted $(D_k)^{\rho_s}$ therefore is equivalent to a circuit of rank at most δ_k , where δ_k is the unique ordinal such that $\beta_s + \delta_k = \text{rank}(D_k)$. In particular, we see that $(D_s)^{\rho_s}$ is equivalent to a finite circuit.

Each restricted subtree $(S_{ij\ell})^{\rho_s}$ is equivalent to a circuit with rank $\gamma_{ij\ell}$, with $\beta_s + \gamma_{ij\ell} = \text{rank}(S_{ij\ell})$. Fix $i, j \in \mathbb{N}$. Then there exists some $S_{ij\ell}$ with $\text{rank}(D_k) < \text{rank}(S_{ij\ell})$, and thus

$$\beta_s + \delta_k = \text{rank}(D_k) < \sup_{\ell \in \mathbb{N}} (\text{rank}(S_{ij\ell})) = \beta_s + \sup_{\ell \in \mathbb{N}} \gamma_{ij\ell}.$$

This means that $\delta_k < \sup_{\ell \in \mathbb{N}} \gamma_{ij\ell}$ by right monotonicity, so the invariant is maintained after each step.

After all stages are complete, we can suppress or preserve any untouched subtrees to obtain a restriction $\tilde{\rho}$. We then have that each $(T_{ij})^{\tilde{\rho}}$ is equivalent to a full rank-1 tree. Since each $(D_k)^{\tilde{\rho}}$ is equivalent to a finite circuit, each $(C_n)^{\tilde{\rho}}$ is equivalent to a Σ_1 - or Π_1 -circuit. Thus we can obtain our desired restriction ρ from the $\alpha = 1$ case to complete the induction in the limit case and prove the theorem. $\square$

4 Consequences of the restriction lemma

Along with being interesting in its own right, the restriction lemma is a powerful tool for proving lower bounds on Borel rank. We demonstrate this fact by providing new proofs for several classical results in descriptive set theory.

4.1 The Borel hierarchy theorem

Theorem 4.1 (Borel hierarchy). *We have for every $1 \leq \gamma < \omega_1$ that $\Sigma_\gamma^0 \neq \Pi_\gamma^0$.*

This theorem is a cornerstone result of descriptive set theory which, along with reductions to complete sets, is a key piece of machinery for proving lower bounds on Borel rank. The usual proof requires diagonalizing against universal sets constructed for each class Σ_γ^0 and Π_γ^0 . The restriction lemma gives us a purely combinatorial proof:

Proof. It suffices to prove the claim when γ is a successor ordinal, so write $\gamma = \beta + 1$. Let $g: 2^\mathbb{N} \rightarrow \{0, 1\}$ be a γ -tree function. Since $g^{-1}(\{1\}) \in \Sigma_\gamma^0 \cup \Pi_\gamma^0$, it suffices to show that $g^{-1}(\{1\})$ is not in $\Delta_\gamma^0 = \Delta_{\beta+1}^0$. Indeed, if this were the case, then g would be $\Sigma_{\beta+1}^0$ -measurable. Applying Theorem 1.2 with $\alpha = 1$ would then give us a restriction ρ with g^ρ simultaneously constant and also a 1-tree function, which is a contradiction. $\square$

Remark 4.2. This result also implies the Borel hierarchy theorem in any uncountable Polish space X , since any such space contains a homeomorphic copy K of the Cantor space. The image of a set which is $\Sigma_\gamma^0 \setminus \Pi_\gamma^0$ in $2^\mathbb{N}$ under a homeomorphism $\varphi: 2^\mathbb{N} \rightarrow K$ can easily be shown to be $\Sigma_\gamma^0 \setminus \Pi_\gamma^0$ in X .

4.2 Ramsey-theoretic results

The restriction lemma allows us to prove some Ramsey-like properties of Borel functions.

Theorem 4.3. *For any Borel function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$, there exists a restriction ρ on $2^\mathbb{N}$ preserving infinitely many variables such that f^ρ is continuous.*

Proof. Fix a Borel function f , and choose $\beta < \omega_1$ such that f is $\Sigma_{\beta+1}^0$ -measurable. Let $g: 2^\mathbb{N} \rightarrow \{0, 1\}$ be a $(\beta + 1)$ -tree function. Theorem 1.1 (with $\alpha = 1$) gives us a restriction ρ

such that f^ρ is continuous while g^ρ is a 1-tree function. As g^ρ depends on infinitely many variables, ρ must preserve infinitely many variables. $\square$

Remark 4.4. The domain of any restricted function is a subcube of the Cantor space. Thus the above result can be interpreted as saying that, if $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ is Borel, then $f|_C$ is continuous for some infinite subcube $C \subseteq 2^\mathbb{N}$.

Remark 4.5. There are other proofs of Theorem 4.3. One such approach is to use the facts that any Borel function $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ is continuous on a comeager subset of $2^\mathbb{N}$ and that every comeager subset of $2^\mathbb{N}$ contains an infinite subcube. A simple argument involving the Galvin-Prikry theorem [GP73] also gives us Theorem 4.3. Both of these alternative approaches, however, rely on either topological machinery or heavy theory.

We also obtain the following Ramsey-like result from the constant restriction lemma:

Theorem 4.6. *For any Borel function $f: 2^\mathbb{N} \rightarrow \{0,1\}$, there exists a restriction ρ on $2^\mathbb{N}$ preserving infinitely many variables such that f^ρ is constant.*

Proof. Choose $\beta < \omega_1$ such that f is $\Sigma_{\beta+1}^0$ -measurable, and let $g: 2^\mathbb{N} \rightarrow \{0,1\}$ be a $(\beta+1)$ -tree function. Use Theorem 1.2 (with $\alpha = 1$) to obtain a restriction ρ such that f^ρ is constant while g^ρ is a 1-tree function, which implies that ρ preserves infinitely many variables. $\square$

Remark 4.7. These results are much weaker than the restriction lemma, since they “forget” the precise structure of ρ . In each proof, we only use the fact that ρ preserves infinitely many variables.

4.3 Infinite parity

An *infinite parity function* is a function $f: 2^\mathbb{N} \rightarrow \{0,1\}$ such that $f(x) \neq f(y)$ whenever $x_i \neq y_i$ for exactly one $i \in \mathbb{N}$. It is a classical result in descriptive set theory that no infinite parity function is Borel. Our argument is perhaps the most similar to Furst, Saxe, and Sipser’s proof of the finitary analog: that parity is not in AC^0 [FSS81].

Corollary 4.8. *No infinite parity function is Borel.*

Proof. Fix a Borel function f , and use Theorem 4.6 to obtain a restriction ρ preserving infinitely many variables for which f^ρ is constant. For any infinite parity function g , the restricted function g^ρ is another infinite parity function, which is not constant. Thus $f^\rho \neq g^\rho$, which implies that $f \neq g$. $\square$

5 Generalization to non-trees

Theorem 1.1 tells us that, if $f: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ is $\Sigma_{\beta+1}^0$ -measurable and $A \subseteq 2^\mathbb{N}$ is a $\Sigma_{\beta+\alpha}^0$ - or $\Pi_{\beta+\alpha}^0$ -complete set computed by a tree function, then there is a subcube $C \subseteq 2^\mathbb{N}$ such that $f|_C$ is continuous and $A \cap C$ is Σ_α^0 - or Π_α^0 -hard. It is a natural question whether we can

take A to be any $\Sigma_{\beta+\alpha}^0$ - or $\Pi_{\beta+\alpha}^0$ -complete set, not necessarily computed by a tree function. The answer is yes if we allow for C to be any closed set rather than just a subcube:

Theorem 5.1. *Fix ordinals $\alpha, \beta < \omega_1$. Then for any $\Sigma_{\beta+1}^0$ -measurable function $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ and $\Sigma_{\beta+\alpha}^0$ - or $\Pi_{\beta+\alpha}^0$ -complete set A , there exists a closed set $C \subseteq 2^{\mathbb{N}}$ such that $f|_C$ is continuous and $A \cap C$ is Σ_{α}^0 - or Π_{α}^0 -hard.*

Proof. Fix a $(\beta + \alpha)$ -tree function $g: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ computing a set in the appropriate class $\Sigma_{\beta+\alpha}^0$ or $\Pi_{\beta+\alpha}^0$ so that there exists a continuous reduction $\varphi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ with $g(x) = 1 \iff \varphi(x) \in A$. The function $f \circ \varphi$ is $\Sigma_{\beta+1}^0$ -measurable, so Theorem 1.1 gives us a restriction ρ such that $(f \circ \varphi)^\rho$ is continuous and g^ρ is computed by an α -tree function. Let C' denote the subcube defined by ρ , and set $C = \varphi(C')$. Then C is compact and hence closed. Because $(f \circ \varphi)|_{C'}$ is continuous, φ is continuous, and C' is compact, a standard topological argument tells us that $f|_C$ is continuous (or one can simply note that φ is a quotient map). Finally, since $\varphi|_{C'}$ is a reduction from the Σ_{α}^0 - or Π_{α}^0 -hard set computed by g^ρ to $A \cap C$, the set $A \cap C$ is Σ_{α}^0 - or Π_{α}^0 -hard. $\square$

Applying a similar argument using Theorem 1.2 gives us the following weaker result:

Theorem 5.2. *Fix ordinals $\alpha, \beta < \omega_1$ such that $\alpha \geq 1$. Then for any $\Sigma_{\beta+\alpha}^0$ - or $\Pi_{\beta+\alpha}^0$ -complete set A and set $B \in \Delta_{\beta+1}^0$, there exists a closed set $C \subseteq 2^{\mathbb{N}}$ such that $A \cap C$ is Σ_{α}^0 - or Π_{α}^0 -hard and $B \cap C$ is either $\emptyset$ or C .*

Remark 5.3. One can also prove these results classically using a refinement of topology.

5.1 Limitations in the infinite setting

The following counterexample tells us that this relaxation from subcubes to closed sets is necessary when considering Σ_{γ}^0 - or Π_{γ}^0 -complete sets which are not computed by tree functions, even for even the weaker result in Theorem 5.2.

Theorem 5.4. *For any countable ordinal γ , there exist Borel functions $f, g: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ such that $f^{-1}(\{1\})$ is open, g computes a Σ_{γ}^0 - or Π_{γ}^0 -hard set, and for any restriction ρ such that f^ρ is constant, the function g^ρ is also constant.*

Proof. We implement the parity gate $\oplus$ on two bits using the formula $x_0 \oplus x_1 = (\bar{x}_0 \wedge x_1) \vee (x_0 \wedge \bar{x}_1)$ and define the functions $d, h: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ by $d(x) = x_0 x_0 x_1 x_1 x_2 x_2 \dots$ and $h(x) = x_0 x_2 x_4 \dots$. Let $\tau: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ be a γ -tree function. We then set

$$f(x) = \bigvee_{i \in \mathbb{N}} x_{2i} \oplus x_{2i+1} \quad \text{and} \quad g(x) = f(x) \vee \tau(h(x)).$$

The function f is computed by a Σ_1 -circuit, so $f^{-1}(\{1\})$ is open. We also have that $\tau = g \circ d$, and since d is a continuous reduction from the Σ_{γ}^0 - or Π_{γ}^0 -complete set computed by τ to $g^{-1}(\{1\})$, this means that g computes a Σ_{γ}^0 - or Π_{γ}^0 -hard set. Finally, suppose ρ is a restriction for which f^ρ is constant. If f^ρ is constantly 1, then g^ρ is constantly 1. If f^ρ is constantly 0, then ρ must fix every x_i , implying that g^ρ is constant. $\square$

We can also set $g(x) = f(x) \vee p(h(x))$ for some infinite parity function p to obtain the following related counterexample:

Theorem 5.5. *There exists a Borel function $f: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ computing an open set and function $g: 2^{\mathbb{N}} \rightarrow \{0, 1\}$ which is not Borel such that, for any restriction ρ such that f^ρ is constant, the function g^ρ is also constant.*

Remark 5.6. Continuous reductions allow us to circumvent the difficulty of working with sets not computed by tree functions, but they destroy the combinatorial structure of restrictions. These counterexamples tell us that this is a necessary trade-off.

5.2 Limitations for constant-depth circuit families

Let AC_d^0 denote the class of problems computable by depth- d polynomial-sized circuit families $(C_n)_{n \in \mathbb{N}}$, and set $\text{AC}^0 = \bigcup_{d \in \mathbb{N}} \text{AC}_d^0$. The proof of Theorem 5.4 carries over seamlessly to this setting, and using Sipser's result in [Sip83] that the finite analogs of the infinite tree functions of depth $d + 1$ are not in AC_d^0 gives us the following result:

Theorem 5.7. *For any $d \geq 2$, there exist functions $f_n, g_n: \{0, 1\}^n \rightarrow \{0, 1\}$ for $n \in \mathbb{N}$ such that $\{f_n\}_{n \in \mathbb{N}} \in \text{AC}_2^0$, $\{g_n\}_{n \in \mathbb{N}} \in \text{AC}_d^0 \setminus \text{AC}_{d-1}^0$, and for every $n \in \mathbb{N}$ and restriction ρ such that f_n^ρ is constant, the function g_n^ρ is also constant.*

Following Theorem 5.5 instead, we get the following:

Theorem 5.8. *There exist functions $f_n, g_n: \{0, 1\}^n \rightarrow \{0, 1\}$ for $n \in \mathbb{N}$ such that $\{f_n\}_{n \in \mathbb{N}} \in \text{AC}_2^0$, $\{g_n\}_{n \in \mathbb{N}} \notin \text{AC}^0$, and for every $n \in \mathbb{N}$ and restriction ρ such that f_n^ρ is constant, the function g_n^ρ is also constant.*

Restrictions are certainly a powerful tool for proving the limitations of AC^0 . Just as in the infinite case, though, our counterexamples show that they are not universally applicable for proving lower bounds on arbitrary functions.

6 Sharpness of the lemma

When applying the restriction lemma to reduce the complexity of a function f by β , we also reduce the complexity of some $(\beta + \alpha)$ -tree function g by β . A natural question is whether it is possible to reduce the complexity of f without reducing the complexity of g , and surprisingly, the answer is yes in certain instances. In this section, we show that the full restriction is sharp, while the constant restriction lemma is not.

6.1 Optimality of the full restriction lemma

The following result tells us that Theorem 1.1 is completely sharp.

Theorem 6.1. *For any ordinals $\alpha, \beta < \omega_1$ and $(\beta + \alpha)$ -tree function $g: 2^{\mathbb{N}} \rightarrow \{0, 1\}$, there is a $\Sigma_{\beta+1}^0$ -measurable function $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that, for any restriction ρ such that f^ρ is continuous, the function g^ρ computes a Borel set of rank at most α .*

Proof. Let C be an alternating tree of rank $(\beta + \alpha)$, and let $\mathcal{F} = \{C_0, C_1, \dots\}$ denote the family of rank- β subcircuits of C . Define $f: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that $f(x)_i = C_i(x)$. Lemma 2.23 tells us that f is $\Sigma_{\beta+1}^0$ -measurable. If ρ is a restriction such that f^ρ is continuous, then Corollary 2.24 tells us that each C_i^ρ is equivalent to a finite circuit. Thus for this restriction ρ , the function g^ρ is computed by a Σ_α - or Π_α -circuit. Thus g^ρ computes a set of rank at most α . $\square$

Remark 6.2. Theorem 1.1 tells us that ρ may always be chosen so that f^ρ is continuous and g^ρ computes a Σ_α^0 - or Π_α^0 -complete set, so both results are optimal for certain choices of f .

6.2 Beating the constant restriction lemma

While the full restriction lemma can be viewed as reducing the rank of countably many circuits (each coordinate of f is computed by a $\Sigma_{\beta+1}$ - and $\Pi_{\beta+1}$ -circuit), Theorem 1.2 only reduces the rank of a single circuit. In this way, the constant restriction lemma is weaker than Theorem 1.1. In fact, it turns out that the constant restriction lemma is not sharp. For example, the constant restriction lemma tells us that for any Σ_1 -circuit C and 2-tree function g , there exists a restriction ρ such that g^ρ is a 1-tree function and C^ρ is constant. However, we can actually choose ρ such that g^ρ is a 2-tree function and C^ρ is constant. This is because one can always force C to be constant by setting finitely many bits, and ρ can be obtained by then suppressing finitely many rank-1 subcircuits in a full alternating rank-2 tree computing g . Using this observation in conjunction with Lemma 3.1, one can obtain a similar rank-1 improvement over Theorem 1.2 for larger values of α and β .

It is therefore clear that Theorem 1.2 is not sharp. In fact, it is an open problem whether forcing C^ρ to be constant requires the complexity of an appropriate tree function (with rank larger than that of C) to drop at all. We pose this question more precisely as follows:

Conjecture 6.3. Suppose $\alpha < \omega_1$, a circuit C has rank strictly less than α , and g is an α -tree function. Then there exists a restriction ρ such that C^ρ is constant and g^ρ is still an α -tree function.

There are a few reasons to suspect that the conjecture holds, and we present two partial results in this direction. The first is that such a restriction can always be found when $\alpha \leq 3$. We prove this via an optimized version of the construction we used to prove the $\alpha = 1$ case of Lemma 3.1.

Theorem 6.4. *Conjecture 6.3 holds whenever $\alpha \leq 3$.*

Proof. The $\alpha = 1$ case is trivial, and we already addressed the $\alpha = 2$ case. It therefore suffices to prove that, whenever C is a Π_2 -circuit and g is a 3-tree function computing a set

in Σ_3^0 , there exists a restriction ρ for which C^ρ is constant and g^ρ is a 3-tree function (the other cases are analogous).

To this end, relabel the input variables $x_0, x_1, \dots$ so that $g(x) = \bigvee_{i \in \mathbb{N}} \bigwedge_{j \in \mathbb{N}} \bigvee_{k \in \mathbb{N}} x_{ijk}$. We call the collection $\{x_{ijk} : k \in \mathbb{N}\}$ of variables the (i, j) -th row. Note that any partial restriction setting variables in only finitely many rows can be extended to a restriction ρ for which g^ρ is a 3-tree function. Call a row *starred* if it contains at least one preserved variable. Write C as the conjunction of Σ_1 -circuits D_n for $n \in \mathbb{N}$. Define the *queue sequence* $(i_n, \ell_n)_{n \in \mathbb{N}}$ so that every $(i, \ell) \in \mathbb{N}^2$ appears infinitely many times in the sequence.

We are done if there exists a restriction ρ such that g^ρ is a 3-tree function and any D_n^ρ is identically 0, so suppose this is not the case. We will construct a restriction ρ such that g^ρ is a 3-tree function and every D_n^ρ is identically 1. We define ρ in stages, assigning one $*$ at each stage and ensuring that after the n -th stage, the circuit D_n is forced to 1. As we construct ρ , we also define *label functions* $J_i : \mathbb{N} \rightarrow \mathbb{N}$. Each J_i will be an injective partial function at finite stages of the construction and a total injective function in the limit. We will ensure that at the end of each stage, the (i, j) -th row is starred if and only if j lies in the image of J_i . In particular, we do the following for $n = 0, 1, 2, \dots$:

1. If $J_{i_n}(\ell_n)$ is not yet defined, choose some $j \in \mathbb{N}$ not in the range of J_{i_n} such that ρ has not yet been defined on any variables in the (i_n, j) -th row. Define $J_{i_n}(\ell_n) = j$.
2. Preserve an additional variable in the $(i_n, J_{i_n}(\ell_n))$ -th row.
3. For each of the 2^{n+1} possible settings of the preserved variables to 0 or 1, do the following:
 - (a) Under this setting, it must be possible to force D_n to 1 while assigning only 0's in starred rows (otherwise, setting all remaining variables in the starred rows to 0 would force D_n to 0, and we could extend this partial restriction to some ρ with g^ρ being a 3-tree function and D_n^ρ being identically 0). Thus one of the Σ_0/Π_0 -children of D_n can be forced to 1 by setting finitely many variables, assigning only 0's in starred rows.
 - (b) For each of the finitely many unstarred rows containing a newly assigned variable, assign an additional 1 in that row so that the row is suppressed.

Let $\tilde{\rho}$ denote the resulting partial restriction, and note that $\tilde{\rho}$ forces every $D_n^{\tilde{\rho}}$ to 1. We also see that every starred row contains only 0's and $*$'s, and that each starred row contains infinitely many stars. For each fixed i , the (i, j) -th row is starred for infinitely many j , and this means that by extending $\tilde{\rho}$ to a restriction ρ suppressing all other rows, the restricted function g^ρ is a 3-tree function. $\square$

The second partial result is that the claim holds when the rigid subcubes induced by restrictions are replaced by closed sets (this is the same relaxation we made in Section 5). The argument was found by Slawomir Solecki, and it relies heavily on a Hurewicz-type result proven by Louveau and Saint Raymond in [LS87].

Theorem 6.5. Fix $\alpha < \omega_1$ and $A, B \subseteq 2^{\mathbb{N}}$ such that A is Π_α^0 -complete and $\text{rank}(B) < \alpha$. Then there exists a closed set $C \subseteq 2^{\mathbb{N}}$ such that $A \cap C$ is Π_α^0 -hard and $B \cap C$ is equal to either $\emptyset$ or C .

Proof. Because $A \notin \Sigma_\alpha^0$, we must have that either $A \cap B \notin \Sigma_\alpha^0$ or $A \setminus B \notin \Sigma_\alpha^0$. We consider each of these cases separately.

We first suppose that $A \cap B \notin \Sigma_\alpha^0$. We can assume that $\alpha \geq 3$ since the $\alpha \leq 2$ cases are trivial, and this allows us to apply Louveau and Saint Raymond’s theorem in [LS87] to the sets $A \cap B$ and $B \setminus A$. The theorem tells us that either there exists a Σ_α^0 -set S with $A \cap B \subseteq S \subseteq A \cup B^c$ or there is a continuous map $\varphi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that $\varphi(H_\alpha) \subseteq A$ and $\varphi(H_\alpha^c) \subseteq B \setminus A$, where H_α is some Π_α^0 -complete subset of $2^{\mathbb{N}}$. The first alternative cannot hold, since it would imply that $A \cap B = S \cap B \in \Sigma_\alpha^0$. Thus the second alternative holds, and $C = \varphi(2^{\mathbb{N}})$ is closed and entirely contained in B . Because φ is a continuous reduction from H_α to $A \cap C$, the set $A \cap C$ is Π_α^0 -hard.

Now suppose that $A \setminus B \notin \Sigma_\alpha^0$. Louveau and Saint Raymond’s theorem applied to $A \setminus B$ and $(A \cup B)^c$ tells us that either there exists a Σ_α^0 -set S with $A \setminus B \subseteq S \subseteq A \cup B$ or there is a continuous map $\varphi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that $\varphi(H_\alpha) \subseteq (A \setminus B)$ and $\varphi(H_\alpha^c) \subseteq (A \cup B)^c$. The first alternative again cannot hold, since we would then have that $A \setminus B = S \setminus B \in \Sigma_\alpha^0$. Thus the second alternative holds, and we see that $C = \varphi(2^{\mathbb{N}})$ is closed and disjoint from B . Because φ is a continuous reduction from H_α to $A \cap C$, the set $A \cap C$ is again Π_α^0 -hard. $\square$

Remark 6.6. This argument tells us that, if Conjecture 6.3 fails, it must be due to the precise combinatorial structure of restrictions (as was the case for Theorem 5.2).

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